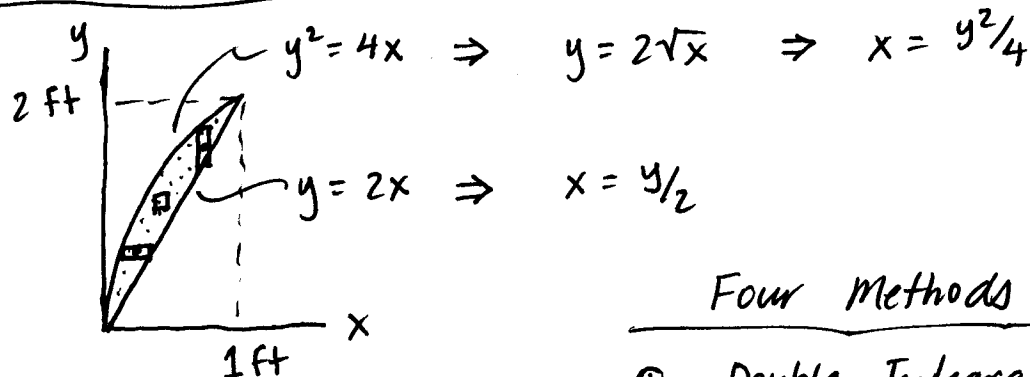


Centroids by Integration

Example Problem: Area bounded by two lines

Find area centroid $(\bar{x}, \bar{y})$ by four methods



Four Methods:

① Double Integral: $dA = dy dx$

$$\tilde{x} = x, \quad \tilde{y} = y$$

$$\bar{x} \text{ numerator: } = \int_0^1 \int_{2x}^{2\sqrt{x}} x dy dx$$

(dy from $2x \rightarrow 2\sqrt{x}$) ↗

② Double Integral: $dA = dx dy$

$$\tilde{x} = x, \quad \tilde{y} = y$$

$$\bar{x} \text{ numerator: } = \int_0^2 \int_{y^2/4}^{y/2} x dx dy$$

(dx from $y^2/4 \rightarrow y/2$) ↗

③ Single Integral (Vertical Element)

$$dA = (2\sqrt{x} - 2x) dx$$

$$\begin{matrix} 2\sqrt{x} = y_{\text{top}} \\ 2x = y_{\text{bot}} \end{matrix}$$

$$\rightarrow 1 \leftarrow dx$$

$$\tilde{x} = x, \quad \tilde{y} = \frac{y_{\text{bot}} + y_{\text{top}}}{2} = \left[\frac{2x + 2\sqrt{x}}{2} \right]$$

$$\bar{x} \text{ numerator: } = \int_0^1 x (2\sqrt{x} - 2x) dx$$

④ Single Integral (Horiz Element)

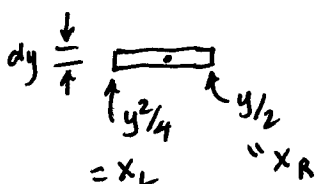
$$dA = (y/2 - y^2/4) dy$$

$$\tilde{y} = y$$

$$\tilde{x} = \left(\frac{x_L + x_R}{2} \right) = \frac{1}{2} \left(y^2/4 + y/2 \right)$$

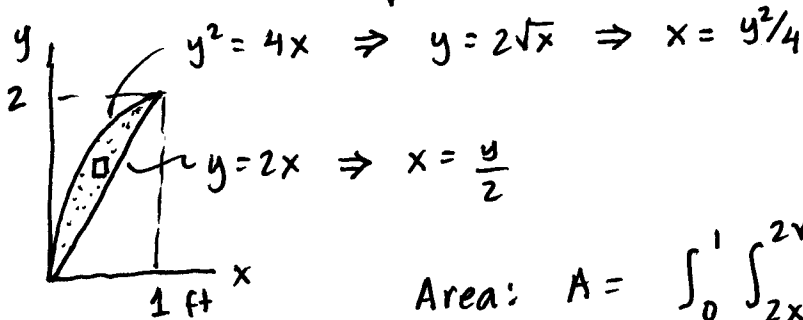
$\bar{x}$ numerator

$$= \int_0^2 \underbrace{\frac{1}{2} \left(\frac{y^2}{4} + \frac{y}{2} \right)}_{\tilde{x}} \underbrace{\left(\frac{y}{2} - \frac{y^2}{4} \right)}_{dA} dy$$



2 Lines Ex Prob (cont'd)

① Double Integral $dA = dy dx$ $\tilde{x} = x, \tilde{y} = y$



$$\begin{cases} dy \text{ limits: } 2x \rightarrow 2\sqrt{x} \\ dx \text{ limits: } 0 \rightarrow 1 \end{cases}$$

$$\begin{aligned} \text{Area: } A &= \int_0^1 \int_{2x}^{2\sqrt{x}} dy dx = \int_0^1 2(\sqrt{x} - x) dx \\ &= 2\left(\frac{2}{3}x^{3/2} - \frac{1}{2}x^2\right)_0^1 = 2\left(\frac{2}{3} - \frac{1}{2}\right) = \left(\frac{4}{6} - \frac{3}{6}\right) \\ &= 2\left(\frac{1}{6}\right) = \frac{2}{6} = \boxed{\frac{1}{3} = 0.333 \text{ ft}^2 = A} \end{aligned}$$

$$\text{TI-89: } \int(\int(1, y, 2*x, 2*x^{.5}), x, 0, 1) = .333 \text{ ft}^2$$

$$\begin{aligned} \bar{x} \text{ Numerator: } &= \int_0^1 \int_{2x}^{2\sqrt{x}} x dy dx = \int_0^1 2x(\sqrt{x} - x) dx \\ &= \int_0^1 2(x^{3/2} - x^2) dx = 2\left(\frac{2}{5}x^{5/2} - \frac{1}{3}x^3\right)_0^1 \\ &= 2\left(\frac{2}{5} - \frac{1}{3}\right) = 2\left(\frac{6}{15} - \frac{5}{15}\right) = \boxed{\frac{2}{15} \text{ ft}^3} \end{aligned}$$

$$\text{TI-89: } \int(\int(x, y, 2*x, 2*x^{.5}), x, 0, 1) = \boxed{.1333 \text{ ft}^3 = \frac{2}{15}}$$

$$\bar{x} = \frac{2/15}{1/3} = \frac{6}{15} = \boxed{\frac{2}{5} = 0.40 \text{ ft} = \bar{x}}$$

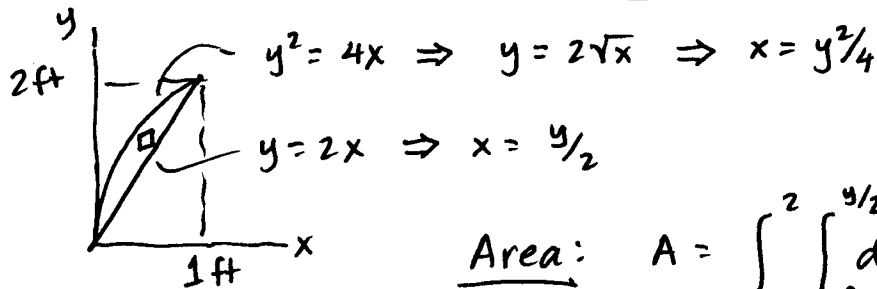
$$\begin{aligned} \bar{y} \text{ Numerator} &= \int_0^1 \int_{2x}^{2\sqrt{x}} y dy dx = \int_0^1 \left.\frac{1}{2}y^2\right|_{2x}^{2\sqrt{x}} dx \\ &= \int_0^1 \frac{1}{2}(4x - 4x^2) dx = \int_0^1 2(x - x^2) dx = 2\left(\frac{1}{2}x^2 - \frac{1}{3}x^3\right)_0^1 \\ &= 2\left(\frac{1}{2} - \frac{1}{3}\right) = 2\left(\frac{3}{6} - \frac{2}{6}\right) = 2\left(\frac{1}{6}\right) = \frac{2}{6} = \frac{1}{3} \text{ ft}^3 = .333 \text{ ft}^3 \end{aligned}$$

$$\text{TI-89: } \int(\int(y, y, 2*x, 2*x^{.5}), x, 0, 1) = .333 \text{ ft}^3$$

$$\bar{y} \text{ centroid: } \bar{y} = \frac{1/3}{1/3} = \boxed{1.0 \text{ ft} = \bar{y}}$$

2 Lines Ex Prob (cont'd)

② Double Integral: $\boxed{dA = dx dy}$ $\tilde{x} = x, \tilde{y} = y$



$$\begin{aligned}\text{Area: } A &= \int_0^2 \int_{y^2/4}^{y/2} dx dy = \int_0^2 \left(\frac{y}{2} - \frac{y^2}{4} \right) dy \\ &= \left(\frac{1}{4} y^2 - \frac{1}{12} y^3 \right)_0^2 = \left(\frac{4}{4} - \frac{8}{12} \right) = \left(1 - \frac{2}{3} \right) = \boxed{\frac{1}{3} = .333 \text{ ft}^2}\end{aligned}$$

$$\text{TI-89: } \int \left(\int (1, x, .25 * y^2, .5 * y), y, 0, 2 \right) = .333 \text{ ft}^2$$

$$\begin{aligned}\bar{x} \text{ numerator: } &= \int_0^2 \int_{y^2/4}^{y/2} x dx dy = \int_0^2 \frac{1}{2} x^2 \Big|_{y^2/4}^{y/2} dy \\ &= \int_0^2 \frac{1}{2} \left(\frac{y^2}{4} - \frac{y^4}{16} \right) dy = \int_0^2 \frac{1}{8} \left(y^2 - \frac{y^4}{4} \right) dy = \frac{1}{8} \left(\frac{1}{3} y^3 - \frac{1}{20} y^5 \right)_0^2 \\ &= \frac{1}{8} \left(\frac{8}{3} - \frac{32}{20} \right) = \left(\frac{1}{3} - \frac{1}{5} \right) = \left(\frac{5}{15} - \frac{3}{15} \right) = \boxed{\frac{2}{15} \text{ ft}^3}\end{aligned}$$

$$\text{TI-89: } \int \left(\int (x, x, .25 * y^2, .5 * y), y, 0, 2 \right) = \boxed{.1333 \text{ ft}^3 = \frac{2}{15}}$$

$$\bar{x} = \frac{2/15}{1/3} = \frac{6}{15} = \frac{2}{5} = \boxed{0.40 \text{ ft} = \bar{x}}$$

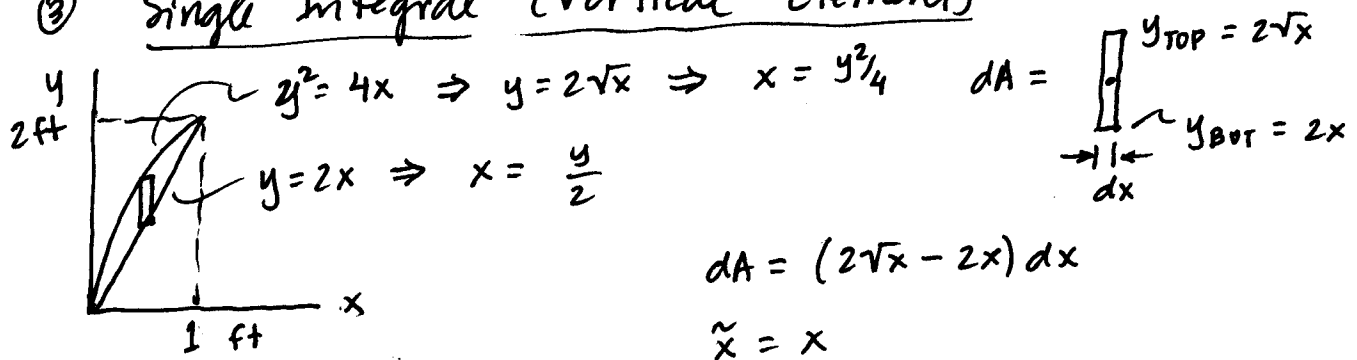
$$\begin{aligned}\bar{y} \text{ numerator: } &= \int_0^2 \int_{y^2/4}^{y/2} y dx dy = \int_0^2 y \left(\frac{y}{2} - \frac{y^2}{4} \right) dy \\ &= \int_0^2 \left(\frac{1}{2} y^2 - \frac{1}{4} y^3 \right) dy = \left(\frac{1}{6} y^3 - \frac{1}{16} y^4 \right)_0^2 = \frac{8}{6} - 1 = \frac{2}{6} = \boxed{\frac{1}{3} \text{ ft}^3}\end{aligned}$$

$$\text{TI-89: } \int \left(\int (y, x, .25 * y^2, .5 * y), y, 0, 2 \right) = .333 \text{ ft}^3$$

$$\bar{y} = \frac{1/3}{1/3} = \boxed{1.0 \text{ ft} = \bar{y}}$$

2 Lines Ex Problem (cont'd)

③ Single Integral (Vertical Element)



$$\text{Area} = \int_0^1 2(x^{1/2} - x) dx = 2\left(\frac{2}{3}x^{3/2} - \frac{1}{2}x^2\right)_0^1 = 2\left(\frac{2}{3} - \frac{1}{2}\right) = 2\left(\frac{4}{6} - \frac{3}{6}\right) = \frac{2}{6} = \frac{1}{3} = \boxed{.333 \text{ ft}^2}$$

$$\text{TI-89: } \int (2*(x^{.5} - x), x, 0, 1) = \boxed{.333 \text{ ft}^2}$$

$$\bar{x} \text{ numerator} = \int_0^1 \underbrace{x}_{\tilde{x}} \underbrace{[2(x^{1/2} - x)]}_{dA} dx = \int_0^1 2(x^{3/2} - x^2) dx$$

$$= 2\left(\frac{2}{5}x^{5/2} - \frac{1}{3}x^3\right)_0^1 = 2\left(\frac{2}{5} - \frac{1}{3}\right) = 2\left(\frac{6}{15} - \frac{5}{15}\right) = \boxed{\frac{2}{15} \text{ ft}^3}$$

$$\text{TI-89: } \int (2*(x^{1.5} - x^2)) = \boxed{.1333 \text{ ft}^3 = \frac{2}{15} \text{ ft}^3}$$

$$\bar{x} = \frac{2/15}{1/3} = \frac{6}{15} = \frac{2}{5} = \boxed{0.40 \text{ ft} = \bar{x}}$$

$$\bar{y} \text{ numerator: } = \int_0^1 \underbrace{(x + x^{1/2})}_{\tilde{y}} \underbrace{[2(x^{1/2} - x)]}_{dA} dx$$

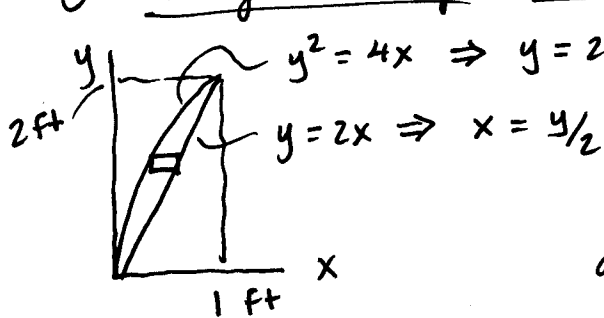
$$= \int_0^1 2(x^{1/2} + x)(x^{1/2} - x) dx = \int_0^1 2(x - x^2) dx = 2\left(\frac{1}{2}x^2 - \frac{1}{3}x^3\right)_0^1 = 2\left(\frac{1}{2} - \frac{1}{3}\right) = 2\left(\frac{3}{6} - \frac{2}{6}\right) = \frac{2}{6} = \frac{1}{3} = .333 \text{ ft}^3$$

$$\text{TI-89: } \int (2*(x - x^2), x, 0, 1) = .333 \text{ ft}^3$$

$$\bar{y} = \frac{1/3}{1/3} = \boxed{1.0 \text{ ft} = \bar{y}}$$

2 Lines Ex Prob (cont'd)

④ Singe Integral (Horiz Element)



$$dy \frac{1}{4} \left\{ \begin{array}{l} x_R \\ x_L \\ = y^2/4 \end{array} \right. \left\{ \begin{array}{l} x_R \\ = y/2 \end{array} \right.$$

$$dA = (x_R - x_L) dy \\ = (y/2 - y^2/4) dy$$

$$\tilde{y} = y \\ \tilde{x} = \frac{x_L + x_R}{2} = \frac{y^2/4 + y/2}{2} = \left(\frac{y^2}{8} + \frac{y}{4} \right)$$

$$\text{Area: } A = \int_0^2 (y/2 - y^2/4) dy = \left(\frac{1}{4} y^2 - \frac{1}{12} y^3 \right)_0^2 = \left(1 - \frac{2}{3} \right) = \frac{1}{3} \text{ ft}^2$$

$$\text{TI-89: } \int (.5 * y - .25 * y^2, y, 0, 2) = \boxed{.333 \text{ ft}^2}$$

$$\begin{aligned} \bar{x} \text{ numerator: } &= \int_0^2 \underbrace{\left(\frac{y^2}{8} + \frac{y}{4} \right)}_{\tilde{x}} \underbrace{\left(\frac{y}{2} - \frac{y^2}{4} \right)}_{dA} dy \\ &= \int_0^2 \frac{1}{2} \left(\frac{y}{2} + \frac{y^2}{4} \right) \left(\frac{y}{2} - \frac{y^2}{4} \right) dy = \int_0^2 \frac{1}{2} \left(\frac{y^2}{4} - \frac{y^4}{16} \right) dy \\ &= \frac{1}{2} \left(\frac{y^3}{12} - \frac{y^5}{80} \right)_0^2 = \frac{1}{2} \left(\frac{8}{12} - \frac{32}{80} \right) = \frac{1}{2} \left(\frac{2}{3} - \frac{4}{10} \right) \\ &= \frac{1}{2} \left(\frac{20}{30} - \frac{12}{30} \right) = \frac{1}{2} \left(\frac{8}{30} \right) = \frac{1}{2} \left(\frac{4}{15} \right) = \frac{2}{15} \text{ ft}^3 \end{aligned}$$

$$\text{TI-89: } \int ((y^2/8 + y/4) * (y/2 - y^2/4), y, 0, 2) = .1333 = \frac{2}{15} \text{ ft}^3$$

$$\bar{x} = \frac{2/15}{1/3} = \frac{6}{15} = \frac{2}{5} = \boxed{0.40 \text{ ft} = \bar{x}}$$

$$\begin{aligned} \bar{y} \text{ numerator: } &= \int_0^2 y \left(\frac{y}{2} - \frac{y^2}{4} \right) dy = \int_0^2 \frac{1}{2} (y^2 - \frac{1}{2} y^3) dy \\ &= \frac{1}{2} \left(\frac{1}{3} y^3 - \frac{1}{8} y^4 \right)_0^2 = \frac{1}{2} \left(\frac{8}{3} - 2 \right) = \frac{4}{3} - 1 = \boxed{\frac{1}{3} \text{ ft}^3} \end{aligned}$$

$$\text{TI-89: } \int (y * (y/2 - y^2/4), y, 0, 2) = .333 \text{ ft}^3$$

$$\bar{y} = \frac{1/3}{1/3} = \boxed{1.0 \text{ ft} = \bar{y}}$$